Name

Mathematics Stage 5 Paths

Expand and Factorise Monic Quadratics

Book 1

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Syllabus Content

Expand binomial products and factorise monic quadratic expressions

- Expand binomial products
- Factorise monic quadratic trinomial expressions
- Solve problems involving expansion and factorisation of algebraic expressions

Expanding Binomial Products using the Rule

Expand Binomial Products using the Rule

1. Multiply each term in the first bracket by each term in the second bracket.
2. Simplify the result.

$$(a+b)(c+d)$$
$$= a(c+d) + b(c+d)$$
$$= ac + ad + bc + bd$$

Shortcut:

$$(a+b)(c+d) = ac + ad + bc + bd$$

Example Expand binomial products

$(x+3)(x+5)$

$$(x+3)(x+5) = x^2 + 5x + 3x + 15$$
$$= x^2 + 8x + 15$$

$(x-4)(x+7)$

$$(x-4)(x+7) = x^2 + 7x - 4x - 28$$
$$= x^2 + 3x - 28$$

Guided Practice

Expand and simplify these expressions.

a $(x+4)(x+2)$

$x^2 + 6x + 8$

b $(x-5)(x-2)$

$x^2 - 7x + 10$

c $(2x-1)(x-6)$

$2x^2 - 13x + 6$

d $(5x-2)(3x+7)$

$15x^2 + 29x - 14$

Key Ideas

1. We can multiply binomials using the rule:

$(a+b)(x+y) = a(\quad) + b(\quad)$

$= \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

1 Expand these expressions.

a	$(x+3)(x-4)$	**b**	$(x-6)(x+2)$	**c**	$(x-2)(x+7)$
	x^2-x-12		$x^2-4x-12$		$x^2+5x-14$
d	$(4x+3)(2x+5)$	**e**	$(2x-3)(3x+5)$	**f**	$(5x+2)(2x-7)$
	$8x^2+26x+15$		$6x^2+x-15$		$10x^2-31x-14$
g	$(3x-2)(6x-5)$	**h**	$(2x+3)(3x-2)$	**i**	$(4x+1)(4x-5)$
	$18x^2-27x+10$		$6x^2+5x-6$		$16x^2-16x-5$
j	$(5x-2)(3x-1)$	**k**	$(3x-2)(2x+1)$	**l**	$(7x-3)(3x-4)$
	$15x^2-11x+2$		$6x^2-x-2$		$21x^2-37x+12$
m	$2(x+3)(x+4)$ $=2(x^2+3x+4x+12)$ $=$	**n**	$-2(x+8)(x+2)$	**o**	$-4(x+9)(x+2)$
	$2x^2+14x+24$		$-2x^2-20x-32$		$-4x^2-44x-72$
p	$3(x+5)(x-3)$	**q**	$-3(a+2)(a-7)$	**r**	$4(a-3)(a-6)$
	$3x^2+6x-45$		$-3a^2+15a+42$		$4a^2-36a+72$
s	$3(2x+3)(2x+5)$	**t**	$6(3x-4)(x+2)$	**u**	$-2(x+4)(3x-7)$
	$12x^2+48x+45$		$18x^2+12x-48$		$-6x^2-10x+56$

2 Compare the expansions of $(x+5)(x+2)$ and $(x-2)(x-5)$.
What's the same and what's different?

The sign of the coefficient of x is different

3 A room in a house with dimensions 4 m by 5 m is to be extended. Both the length and width are to be increased by x m.

a Write an expanded expression for the area of the new room.

$x^2 + 9x + 20$

b If $x = 3$,

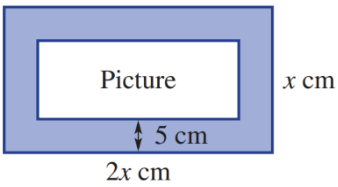
i Find the area of the new room.

$56\ \text{m}^2$

ii By how much the area has increased?

$36\ \text{m}^2$

4 A picture frame 5 cm wide has a length that is twice the width x cm. :



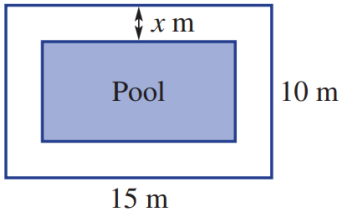
a Find an expression for the total area of the frame and the picture.

$2x^2$

b Find an expression for the area of the picture, in expanded form.

$2x^2 - 30x + 100$

5 The outside edge of a path around a rectangular swimming pool is 15 m long and 10 m wide. The path is x metres wide.



a Find an expression for the area of the pool in expanded form.

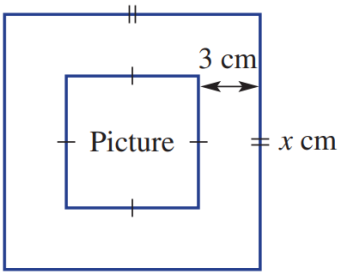
$(10-2x)(15-2x) = 4x^2 - 50x + 150$

b Find the area of the pool if x = 2.

$66\ \text{m}^2$

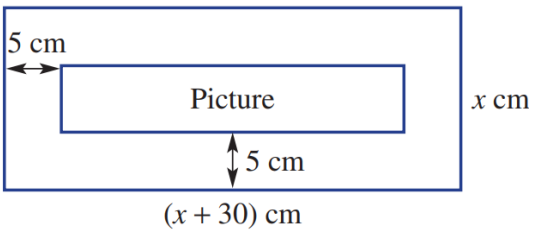
6 Find an expanded expression for the area of the pictures centred in these frames.

a



$x^2 - 12x + 36$

b



$x^2 + 10x - 200$

7 Expand and simplify the following.

a $(x + 4)(x + 3) + x - 4$

$x^2 + 8x + 8$

b $3x^2 + (2x - 1)(x - 2)$

$5x^2 - 5x + 2$

c $(x + 8)(x + 3) + (x + 4)(x + 5)$

$2x^2 + 20x + 44$

d $(y - 7)(y + 4) + (y + 5)(y - 3)$

$2y^2 - y - 43$

e $(x - 5)(x + 5) + (x + 5)(x + 5)$

$2x^2 + 10x$

f $(2a + 3)(a - 5) - (a + 6)(2a + 5)$

$-24a - 45$

g $(x - 3)(x - 4) - (x + 2)(x - 1)$

$-8x + 14$

h $3 - (2x - 9)^2$

$-4x^2 + 36x - 78$

8 Show, by expansion, that $(x + a)(x + b) = x^2 + (a + b)x + ab$

Use this fact to find possible values of a and b for these expansions.

a $(x + a)(x + b) = x^2 + 5x + 6$

$(x + 2)(x + 3)$

b $(x + a)(x + b) = x^2 - 5x + 6$

$(x - 2)(x - 3)$

c $(x + a)(x + b) = x^2 + x - 6$

$(x + 3)(x - 2)$

d $(x + a)(x + b) = x^2 - x - 6$

$(x - 3)(x + 2)$

9 Write the missing terms in these expansions.

a $(x + 2)(x + __) = x^2 + 5x + 6$

3

b $(x + __)(x + 5) = x^2 + 7x + 10$

2

c $(x + 1)(x + __) = x^2 + 7x + __$

6, 6

d $(x + __)(x + 9) = x^2 + 11x + __$

2, 18

e $(x + 3)(x - __) = x^2 + x - __$

2, 6

f $(x - 5)(x + __) = x^2 - 2x - __$

3, 15

g $(x + 1)(__ + 3) = 2x^2 + __ + __$

$2x, 5x, 3$

h $(__ - 4)(3x - 1) = 9x^2 - __ + __$

$3x, 15x, 4$

i $(x + 2)(__ + __) = 7x^2 + __ + 6$

$7x, 3, 17x$

j $(__ - __)(2x - 1) = 6x^2 - __ + 4$

$3x, 4, 2x$

10 Using the area model, or otherwise, expand these expressions.

a $(x + 1)(x^2 + x + 1)$

$x^3 + 2x^2 + 2x + 1$

b $(x - 2)(x^2 - x + 3)$

$x^3 - 3x^2 + 5x - 6$

c $(2x + 1)(2x^2 - x + 4)$

$4x^3 + 7x + 4$

d $(x^2 - x + 1)(x + 3)$

$x^3 + 2x^2 - 2x + 3$

e $(5x^2 - x + 2)(2x - 3)$

$10x^3 - 17x^2 + 7x - 6$

f $(2x^2 - x + 7)(4x - 7)$

$8x^3 - 18x^2 + 35x - 49$

g $(x + 1)(x + 2)(x + 3)$

$x^3 + 6x^2 + 11x + 6$

11 All these expressions are equivalent. Find the values of all the letters, if possible.

$(x + 5)(x + 4)$

$(x + 3)(x + 6) + a$

$(x + 2)(x + 10) + bx$

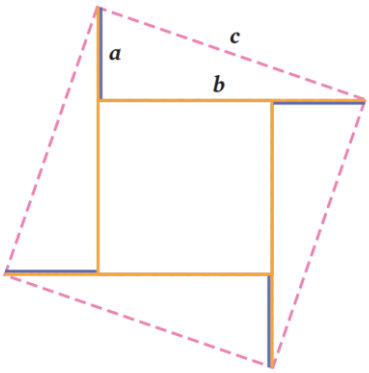
$(x + 5)^2 + cx + d$

$(x + 1)(x + e) + fx + g$

$a = 2, b = -3, c = -1, d = -5$

$e = 1, f = 1, g = 19$ (many answers are possible)

12 Four congruent triangles make a square with side lengths equal to c.



a What is the base and height of each triangle?

a, b

b Find the area of each triangle.

$\frac{ab}{2}$

c What is the side length of the inner square?

$b - a$

d Find the area of the inner square.

$b^2 - 2ab + a^2$

e Write an expression for the combined area of the four triangles and the inner square in terms of a and b.

$a^2 + b^2$

f What theorem have you proven?

$a^2 + b^2 = c^2$

Pythagoras' Theorem

Factorising Monic Quadratic Trinomials

ⓘ Factorising Monic Quadratic Trinomials

1. A **trinomial** is an expression with **three terms**.
2. A **quadratic** expression has highest power x^2.
3. A **monic** quadratic has **1** as the coefficient of x^2.
4. A **monic quadratic trinomial** is an expression of the form $x^2 + bx + c$

🔍 **Identify** monic quadratic trinomials

Determine whether these are monic quadratic trinomials.

a	$x^2 + 3x + 5$	**b**	$2x^2 + 3x + 5$	**c**	$-x^2 + 3x + 5$
d	$x^2 - 3x - 9$	**e**	$8 + 2x + x^2$	**f**	$7 + 2x - x^2$
g	$5x^2 - 7x + 3$	**h**	$x^3 + x^2 + 12x + 7$	**i**	$(x + 3)(x - 8)$
j	$(x - 8)(x - 2)$	**k**	$(2x + 3)(x + 5)$	**l**	$(8 - x)(x + 3)$

Rule for expanding $(x+\alpha)(x+\beta)$

a $(x+5)(x+3)=$	**a** $(x+5)(x+3)=$
b $(x+5)(x+2)=$	**b** $(x+5)(x+2)=$
c $(x+5)(x+0)=$	**c** $(x+5)(x+0)=$
d $(x+5)(x-1)=$	**d** $(x+5)(x-1)=$
e $(x+5)(x-2)=$	**e** $(x+5)(x-2)=$
f $(x+5)(x-3)=$	**f** $(x+5)(x-3)=$

To find the product of two monic binomials $(x+\alpha)(x+\beta)$,

the coefficient of x is …… and the constant term is ……

Complete the following:

g $(x+1)(x-12)=$	**g** $(x+2)(x-12)=$
h $(x+3)(x-20)=$	**h** $(x+7)(x-5)=$
i $(x\quad)(x\quad)=x^2+7x+12$	**i** $(x\quad)(x\quad)=x^2+3x-10$
j $(x\quad)(x\quad)=x^2+17x+70$	**j** $(x\quad)(x\quad)=x^2-2x-24$

Factorising Monic Quadratic Trinomials

To factorise a monic quadratic trinomial: $x^2 + bx + c = 0$:

1. Identify the product: c
2. Identify the sum: b
3. Think of a factor pair, α and β, that has a product of c and sum to b.
4. Factorise $x^2 + bx + c = (x + \alpha)(x + \beta)$.

Example Factorise monic quadratic trinomials

$x^2 - 13x + 36$

1. P: 36
2. S: -13
3. F: $-4, -9$
4. $(x - 4)(x - 9)$

$a^2 + 12a - 28$

1. P: -28
2. S: 12
3. F: $14, -2$
4. $(a + 14)(a - 2)$

Guided Practice

Factorise these expressions.

a $a^2 + 3a + 2$

$(a \quad)(a \quad)$

P: 2

S: 3

F: .....

$(a + 2)(a + 1)$

b $y^2 + 9y + 20$

P: .....

S: .....

F: .....

$(y + 4)(y + 5)$

c $x^2 - 7x + 12$

P: 12

S: -7

F: .....

$(x - 3)(x - 4)$

d $w^2 - 2w - 8$

P: .....

S: .....

F: .....

$(w - 4)(w + 2)$

e $y^2 - 10y + 21$

P: .....

S: .....

F: .....

$(y - 7)(y - 3)$

f $x^2 - 5x - 24$

P: .....

S: .....

F: .....

$(x - 8)(x + 3)$

Key Ideas

1. The expansion of $(x + \alpha)(x + \beta) =$ .....
2. This means that to "un-expand," or factorise, we need to find two numbers that sum to the ..... of x and whose product is the ..... term.

1 Fill in the table.

Binomial Product	Expansion	Factors	Sum	Product
$(x+1)(x+3)$	$x^2+3x+x+3$ $=x^2+4x+3$	1, 3	4	3
$(x+2)(x+7)$	$x^2+9x+14$	2,7	9	14
$(x-3)(x+11)$	$x^2+8x-33$	-3,11	8	-33
$(x+12)(x-5)$	$x^2+7x-60$	12, -5	7	-60
$(x-2)(x-6)$	$x^2-8x+12$	-2, -6	-8	12

2 Fill in the table.

Factor 1	Factor 2	Sum	Product
5	7	12	35
-5	7	2	-35
5	-7	-2	-35
-5	-7	-12	35
3	5	8	15
-3	5	2	-15
3	-5	-2	-15
-3	-5	-8	15
4	6	10	24
-4	6	2	-24
4	-6	-2	-24
-4	-6	-10	24

3 Factorise each expression using the PSF method.

a $x^2 + 3x + 2$

P: ……
S: ……
F: ……

$(x + 1)(x + 2)$

b $x^2 + 7x + 6$

P: ……
S: ……
F: ……

$(x + 6)(x + 1)$

c $x^2 + 8x + 7$

P: ……
S: ……
F: ……

$(x + 7)(x + 1)$

d $x^2 + 6x + 8$

P: ……
S: ……
F: ……

$(x + 4)(x + 2)$

e $x^2 + 3x + 2$

P: ……
S: ……
F: ……

$(x + 2)(x + 1)$

f $x^2 + 7x + 6$

P: ……
S: ……
F: ……

$(x + 6)(x + 1)$

g $x^2 + 10x + 16$

P: ……
S: ……
F: ……

$(x + 8)(x + 2)$

h $x^2 + 9x + 20$

P: ……
S: ……
F: ……

$(x + 4)(x + 5)$

i $x^2 + 3x - 4$

P: ……
S: ……
F: ……

$(x + 4)(x - 1)$

j $x^2 + 4x - 5$

P: ……
S: ……
F: ……

$(x + 5)(x - 1)$

k $x^2 + 2x - 15$

P: ……
S: ……
F: ……

$(x + 5)(x - 3)$

l $x^2 + 3x - 18$

P: ……
S: ……
F: ……

$(x + 6)(x - 3)$

m $x + x^2 - 12$

P: ……
S: ……
F: ……

$(x - 3)(x + 4)$

n $14 + 15x + x^2$

P: ……
S: ……
F: ……

$(x + 14)(x + 1)$

o $x^2 + 24 + 11x$

P: ……
S: ……
F: ……

$(x + 3)(x + 8)$

p $x^2 - 20 + 8x$

P: ……
S: ……
F: ……

$(x + 10)(x - 2)$

4 Write the missing number.

a $(x + \square)(x + 2) = x^2 + 5x + 6$

3

b $(x + \square)(x + 5) = x^2 + 8x + 15$

3

c $(x + 7)(x - \square) = x^2 + 4x - 21$

3

d $(x + 4)(x - \square) = x^2 - 4x - 32$

8

e $(x - 6)(x - \square) = x^2 - 7x + 6$

1

f $(x - \square)(x - 8) = x^2 - 10x + 16$

2

5 Factorise each expression using the PSF method.

a $x^2 - 6x + 5$

$(x - 5)(x - 1)$

b $x^2 - 5x + 4$

$(x - 4)(x - 1)$

c $x^2 - 4x + 4$

$(x - 2)^2$

d $x^2 - 11x + 18$

$(x - 9)(x - 2)$

e $x^2 - 5x + 6$

$(x - 3)(x - 2)$

f $x^2 - 7x - 8$

$(x - 8)(x + 1)$

g $x^2 - 5x - 6$

$(x + 1)(x - 6)$

h $x^2 - 2x - 24$

$(x + 4)(x - 6)$

i $x^2 - x - 12$

$(x - 4)(x + 3)$

j $x^2 - 4x - 12$

$(x - 6)(x + 2)$

k $12 - 8x + x^2$

$(x - 2)(x - 6)$

l $x^2 + 21 - 10x$

$(x - 7)(x - 3)$

m $x^2 - 15 - 2x$

$(x - 5)(x + 3)$

n $-11x + x^2 - 12$

$(x - 12)(x + 1)$

6 Match each quadratic expression to its factored form.

Expanded form	Factored form
1 $x^2 + 14x + 24$	A $(x - 8)(x + 3)$
2 $x^2 + 10x - 24$	B $(x - 6)(x - 4)$
3 $x^2 + 10x + 24$	C $(x + 12)(x - 2)$
4 $x^2 - 10x + 24$	D $(x + 8)(x - 3)$
5 $x^2 - 5x - 24$	E $(x + 12)(x + 2)$
6 $x^2 + 5x - 24$	F $(x + 6)(x + 4)$

1E, 2C, 3F, 4B, 5A, 6D

7 Factorise these expressions. Find product, sum, and factors mentally. Do not write any working out.

a $x^2 + 13x + 36$

$(x + 9)(x + 4)$

b $x^2 - 13x + 36$

$(x - 9)(x - 4)$

c $x^2 - 20x + 36$

$(x - 2)(x - 18)$

d $x^2 + 20x + 36$

$(x + 2)(x + 18)$

e $x^2 + 15x + 36$

$(x + 12)(x + 3)$

f $x^2 - 15x + 36$

$(x - 12)(x - 3)$

g $x^2 + 37x + 36$

$(x + 36)(x + 1)$

h $x^2 - 12x + 36$

$(x - 6)^2$

i $x^2 + 9x - 36$

$(x - 12)(x + 3)$

j $x^2 + 5x - 24$

$(x + 8)(x - 3)$

k $x^2 - 5x - 24$

$(x - 8)(x + 3)$

l $x^2 + 20x + 64$

$(x + 4)(x + 16)$

m $x^2 + 12x - 64$

$(x + 16)(x - 4)$

n $x^2 + 10x + 16$

$(x + 2)(x + 8)$

o $x^2 + 10x + 21$

$(x + 3)(x + 7)$

p $x^2 + 10x + 25$

$(x + 5)^2$

q $x^2 + 6x - 40$

$(x + 10)(x - 4)$

r $x^2 + 3x - 40$

$(x + 8)(x - 5)$

8 Factorise these expressions by first taking out a common factor.

a $2x^2 + 10x + 8$

$= 2(x^2 + 5x + 4)$

$= 2(\quad)(\quad)$

$2(x + 4)(x + 1)$

b $3x^2 + 18x + 24$

$3(x + 4)(x + 2)$

c $2x^2 - 14x - 36$

$2(x - 9)(x + 2)$

d $2x^2 + 2x - 12$

$2(x + 3)(x - 2)$

e $5x^2 - 30x + 40$

$5(x - 4)(x - 2)$

f $2x^2 - 6x - 20$

$2(x - 5)(x + 2)$

g $3x^2 - 33x + 90$

$3(x - 5)(x - 6)$

h $-y^2 + 2y + 15$

$-(y - 5)(y + 3)$

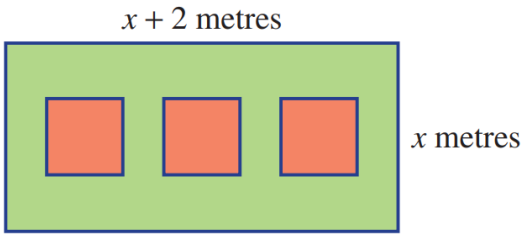
i $-a^2 + 3a + 10$

$-(a - 5)(a + 2)$

j $-4x^2 + 8x - 4$

$-4(x - 1)^2$

9 A backyard, rectangular in area, has a length 2 metres more than its width (x metres).
Inside the rectangle are three square paved areas each of area 5 m^2 as shown.
The remaining area is lawn.
Find an expression for the:



a total backyard area

$x^2 + 2x$

b area of lawn in expanded form

$x^2 + 2x - 15$

c area of lawn in factorised form

$(x + 5)(x - 3)$

10 Factorise $25x^2 + 70x + 48$

Hint: let $a = 5x$

$(5x + 6)(5x + 8)$